



Module 1: Rules & Operating

FCC Privileges · Operating Procedures · Digital Modes
2024–2028 Amateur Extra Question Pool



What We're Building Toward

- 7 modules × 2 hours = 50-question Amateur Extra exam
- Module 1 (today): Rules & Operating — FCC privileges, satellites, digital modes
- Module 2: Propagation & Practices — EME, meteor scatter, test equipment
- Module 3: Electrical Principles — resonance, Q, time constants, phasors
- Module 4: Circuit Components — semiconductors, diodes, digital ICs
- Module 5: Practical Circuits — amplifiers, filters, DSP, oscillators
- Module 6: Signals, Emissions & Safety
- Module 7: Antennas & Transmission Lines — heaviest math, Smith chart
- 50 questions on the exam, 37 to pass (74%) — you already know how to study



The Extra Exam — What to Expect

- 50 multiple-choice questions, randomly drawn from the question pool
- One question from each of 50 groups (E1A through E9H and E0A)
- 37 correct answers to pass = 74%
- No trick questions — four choices, one is always correct
- Exam sessions at the lab — same VE team as Technician and General
- No FCC application fee for upgrade (it's already associated with your license)
- Your call sign doesn't change when you upgrade
- HamStudy is the primary study tool — use it between every session



HamStudy Setup

- Go to hamstudy.org — create or log into your account
- Select the 2024–2028 Amateur Extra question pool
- Share your progress link with your instructor (QR code on handout)
- Between sessions: drill the assigned sections to 80%+ confidence
- HamStudy tracks which questions you miss and drills those harder
- Trust the system — it works the same way it worked for Tech and General
- Target: 80%+ on E1A through E2E before Module 2



Part 1: Extra-Class Frequency Privileges

Subelement E1A — 1 exam question from this group



Where Only Extras Can Transmit

- 80 meters: 3.500–3.525 MHz (CW) and 3.600–3.700 MHz (phone/CW/data)
- 40 meters: 7.000–7.025 MHz (CW)
- 20 meters: 14.000–14.025 MHz (CW) and 14.150–14.175 MHz (phone/CW)
- 15 meters: 21.000–21.025 MHz (CW)
- 10 meters: 28.000–28.025 MHz (CW)
- These are small segments — Extra's reward for doing the harder exam
- General already has most HF — Extra opens the competitive CW segments



Signal Bandwidth at Band Edges

- Your signal extends beyond your carrier frequency — not just on the carrier itself
- USB (upper sideband): signal extends upward from carrier — stay 3 kHz inside upper edge
- LSB (lower sideband): signal extends downward from carrier — stay 3 kHz inside lower edge
- CW: carrier IS the signal center — still has some bandwidth (filter shape)
- FT8 and other digital modes also have real bandwidth — know where you sit
- Transmitting outside the band is an FCC violation even if the carrier is inside



60m, 630m, and 2200m Bands

- 60 meters (5 MHz): channelized operation on 5 specific channel center frequencies
 - → 5332, 5348, 5358.5, 5373, and 5405 kHz
 - → CW must be transmitted at the channel center (not USB-offset like phone)
 - → Maximum power: 100 watts ERP
- 2200 meters (135.7–137.8 kHz): maximum 1 watt EIRP
- 630 meters (472–479 kHz): maximum 5 watts EIRP
- Both LF bands require UTC (Utilities Technology Council) notification
- Wait 30 days after notification before operating — unless UTC clears you earlier
- EIRP = Effective Isotropic Radiated Power (referenced to a perfect isotropic)



Part 2: Station Restrictions

Subelement E1B — 1 exam question from this group



Spurious Emissions & Zoning (E1B)

Spurious Emissions

- Any transmission not part of intended signal
- HF standard: -43 dB below fundamental
- Includes: harmonics, IMD, parasitic oscillations
- Switching amplifiers need output filters
- Your amp must meet this standard

PRB-1 Zoning Preemption

- FCC preemption of local zoning rules
- Local law must 'reasonably accommodate'
- Cannot 'preclude' all amateur communications
- Does NOT eliminate local authority
- Does NOT cover HOA private covenants
- Some states extended similar rules to HOAs



RACES, Antenna Structures & Quiet Zone

- RACES (Radio Amateur Civil Emergency Service):
 - → Requires FCC amateur license AND civil defense certification — both required
 - → May use any frequency authorized to your station during declared emergencies
 - → Limited to 1 hour per week of RACES practice (no limit during actual emergency)
- Antenna structure registration (Part 17 / FAA):
 - → Structures over 200 feet above ground require FAA determination
 - → Structures within certain airport proximity also require notification
 - → FCC ASR database tracks registered structures
- National Radio Quiet Zone: area around Green Bank, WV — restrictions on transmitted power near sensitive observatories



Part 3: Remote/Automatic Control

Subelement E1C — 1 exam question from this group



Remote Control & Control Link Failure

- Remote control: control operator is NOT at the transmitter site
- Control link: the connection between operator and transmitter (internet, RF link, phone)
- If the control link fails: station must automatically stop transmitting within 3 minutes
- The control operator must be able to immediately terminate transmission at any time
- HF FM: modulation index must not exceed 1.0 below 29.7 MHz
- Automatic control: no operator present — station runs itself
- Automatic control is allowed for: beacons, repeaters, RTTY/data gateways
- Automatically controlled data stations may forward third-party traffic on RTTY/data modes



Operating Internationally (CEPT, IARP)

- CEPT License: operate in CEPT member countries without a separate host-country license
 - → CEPT = Conference of Postal and Telecommunications Administrations
 - → Covers most of Europe plus some other countries
 - → You operate under your US license — host country may impose equivalent class limits
- IARP Permit: operate in IARP countries (mainly Americas)
 - → IARP = International Amateur Radio Permit
 - → Required by some Western Hemisphere countries
- Canadian amateurs operating in the US: use Canadian license terms, up to Extra-equivalent
- Check current country-specific rules before any international operation
- STA (Special Temporary Authority): FCC authorization for experimental operations



Part 4: Space Stations & One-Way Comms

Subelement E1D — 1 exam question from this group



Space Station Definitions

- Space station: amateur station located above 50 km altitude
- Telemetry: one-way data transmission FROM space station TO Earth (status, sensor data)
- Telecommand: one-way control transmission FROM Earth TO space station (commanding it)
- Space telecommand stations: only specifically designated amateur stations may transmit to control a space station
- Model craft telecommand: ANY amateur may transmit up to 1 watt to control models — no ID required
- HF bands with space allocations: 40m, 20m, 15m, and 10m
- Balloon stations must transmit ID at regular intervals — standard amateur rules apply
- One-way amateur transmissions are allowed for: beacons, space telemetry, and telecommand only



Part 5: The Volunteer Examiner Program

Subelement E1E — 1 exam question from this group



VE Program Rules

- VEC (Volunteer Examiner Coordinator): organization that coordinates VEs — ARRL, W5YI, Laurel, etc.
- VE (Volunteer Examiner): individual licensed amateur who administers exams
- VE must be accredited (confirmed) by a VEC — not just licensed
- VEs can only be reimbursed for expenses directly related to exam preparation, processing, and administration
- A VE may NOT examine immediate family members or household members
- If an applicant fails an element: VE team RETURNS the application to the applicant — does not keep it
- Fraudulent exam administration by a VE:
 - → Revocation of station license AND suspension of operator license
 - → This is the most severe consequence for VE misconduct



Part 6: Miscellaneous Rules

Subelement E1F — 1 exam question from this group



Miscellaneous Extra-Class Rules (E1F)

- External RF power amplifiers: must meet FCC certification if capable of operating below 144 MHz
- → Prevents use on CB and other services
- Spread spectrum: only authorized above 222 MHz (not on HF)
- Line A: geographic boundary near Canadian border — 420–430 MHz operations restricted near Line A
- Canadian amateurs in the US:
 - → Operate under Canadian license terms
 - → Privileges limited to US equivalent of their Canadian class (up to Extra)
- Auxiliary stations: Technician class and above may serve as control operator
- Mesh networks (AREDN): cannot transmit messages encoded to obscure meaning
- → Encryption to hide content is prohibited on mesh networks just as everywhere else



Part 7: Amateur Satellites

Subelement E2A — 1 exam question from this group



Orbital Mechanics for Ham Radio

- Ascending pass: satellite travels south to north across your location
- Descending pass: satellite travels north to south
- Perigee: closest point of orbit to Earth → least path loss → strongest signal
- Apogee: farthest point → weakest signal, most Doppler shift
- Geostationary: ~35,786 km altitude, satellite appears stationary from Earth
- Keplerian elements: 6 parameters that fully define a satellite's orbit
 - → Used by tracking software (like Gpredict) to predict passes
 - → Updated regularly as orbits drift due to atmospheric drag



Linear Transponders — Key Concept

- A linear transponder receives a range of frequencies on one band, retransmits on another
- Linear transponders are INVERTING:
 - → Upper Sideband uplink becomes Lower Sideband downlink
 - → Signal at the top of the uplink passband appears at the BOTTOM of the downlink passband
- Doppler correction: as satellite approaches, frequencies shift up; as it recedes, they shift down
- Keep your ERP (power × antenna gain) at minimum needed:
 - → High power crushes the transponder's automatic level control
 - → Degrades the downlink signal for EVERYONE using the satellite
- Spin modulation: rotating satellite causes rapid signal variations → use circular polarization



Satellite Mode Designators & Bands

- Mode designators use letters for uplink/downlink bands:
- → Mode U/V: uplink on 70cm (UHF), downlink on 2m (VHF)
- → Mode V/U: uplink on 2m (VHF), downlink on 70cm (UHF)
- → Mode L/U: uplink on 23cm (L band), downlink on 70cm
- L band = 23 cm = 1.2 GHz range
- S band = 13 cm = 2.4 GHz range
- Low-Earth orbit satellites: typical pass duration 8–12 minutes
- Geostationary amateur satellites (like Phase 3 spacecraft): visible continuously



Part 8: Television, Contesting & DX

Subelements E2B–E2C — 1 exam question from this group



Slow-Scan Television (SSTV)

- SSTV transmits still images via audio tones over standard SSB equipment
- Colors are sent SEQUENTIALLY — one color scan per line (not simultaneously)
- Tone frequency encodes pixel brightness: higher frequency = brighter pixel
- VIS (Vertical Interval Signaling) code at the start of each frame:
 - → Identifies the SSTV mode (color system + resolution)
 - → Received automatically by SSTV software to decode correctly
- DRM (Digital Radio Mondiale): uses SSB-capable receiver
- Fast-scan ATV (amateur television): 525 lines NTSC, interlaced scanning
- DVB-T (digital ATV): uses QAM and QPSK modulation



Contest Logging & DX Operating

- Cabrillo format: the standard electronic log format for contest submission
 - → Required by ARRL, CQ Magazine, and most major contest sponsors
- ADIF (Amateur Data Interchange Format): exchange log data between programs
- LoTW (Logbook of the World): ARRL's electronic QSL system
 - → Confirms all contact types (phone, CW, digital) for DXCC, WAS, and other awards
- DX pileup procedure:
 - → Send your FULL call sign once or twice — not just suffix
 - → Listen to how the DX is working: by suffix? by region? simplex or split?
- Split operation: DX transmits one freq, listens on a range 5–10 kHz away
- Remote operation in the US: no '/R' or additional indicator required



Part 9: Digital Modes

Subelements E2D–E2E — 1 exam question from this group



Weak-Signal Digital Modes

- FT8: 15-second TX/RX cycles; 4-tone CPFSK modulation; computer clock must be within ± 1 second
 - → Most popular digital mode in ham radio today
- FT4: 7.5-second cycles; faster paced for contests; less sensitive than FT8
- Q65: designed for EME (Earth-Moon-Earth / moon bounce)
 - → Handles severe Doppler shifting and libration fading
- MSK144: designed for meteor scatter
 - → Uses sporadic E-layer ionization trails from meteors
 - → Works on 28–148 MHz
- FST4: four-tone Gaussian FSK; variable periods (15 seconds to 1800 seconds)
 - → Used on VLF, LF, and MF bands where propagation is very slow
- WSPR: propagation beacons ONLY — not two-way keyboard communication



HF Data Modes & APRS

- PACTOR IV: highest data throughput of any HF digital mode
 - → Used extensively for Winlink email over HF
- PSK31: variable-length (Varicode) character coding
 - → Very narrow bandwidth — second narrowest after FT8
- ALE (Automatic Link Establishment): scans a list of frequencies, selects best one automatically
- APRS (Automatic Packet Reporting System):
 - → Protocol: AX.25 packet radio
 - → Frame type: unnumbered information (UI) frames — connectionless, broadcast
 - → WIDE3-1 digipeater path: relayed by up to 3 digipeaters (counter decrements each hop)
 - → Uses: position reporting, weather data, messages, object tracking



Quick Reference

The numbers and rules that matter



Key Numbers for the Exam (Part 1)

Extra-only 80m CW segment	3.500-3.525 MHz
Extra-only 20m phone segment	14.150-14.175 MHz
60m max power	100W ERP, channelized
2200m max power	1 watt EIRP
630m max power	5 watts EIRP
630m/2200m startup	UTC notification: 30-day wait
Control link failure	Auto-stop within 3 minutes
Spurious emissions (HF)	-43 dB below fundamental
Spread spectrum above	222 MHz
FT8 cycle	15 sec TX/RX; 4-tone CPFSK
MSK144	Meteor scatter, 28-148 MHz
WSPR	One-way beacons only



Key Numbers for the Exam (Part 2)

PACTOR IV	Highest HF data throughput
Cabrillo format	Electronic contest log submission
LoTW	Confirms all contact types for awards
APRS protocol	AX.25, unnumbered information frames
VE cannot examine	Immediate family / household members
VE fraud consequence	Station license revoked + operator license suspended
RACES requirement	FCC license + civil defense certification
PRB-1	Zoning must reasonably accommodate amateur radio
CEPT permit	Operate in CEPT countries on your US license
Satellite transponder	Inverting: USB in, LSB out
Perigee	Closest approach: least path loss
Model craft telecommand	Any amateur, 1W max, no ID required



Checkpoint Quiz 1 / 10

What is the maximum power allowed on the 2200-meter amateur band?

- A. 100 watts PEP
- B. 5 watts EIRP
- C. 1 watt EIRP
- D. 50 watts PEP

Answer: C — 1 watt EIRP (2200m = 135.7–137.8 kHz)



Checkpoint Quiz 2 / 10

Which of the following describes the operation of a linear transponder in an amateur satellite?

- A. Amplifies signals and retransmits on the same frequency
- B. Receives signals and retransmits them in an inverted passband
- C. Converts digital signals to analog and retransmits them
- D. Demodulates and remodulates signals to reduce noise

Answer: B — Inverting passband: USB in = LSB out; signal position reversed



Checkpoint Quiz 3 / 10

What must happen before operating on the 630-meter band?

- A. Notify the ARRL at least 30 days before operating
- B. Notify the Utilities Technology Council and wait 30 days
- C. File an STA request with the FCC
- D. Register with a VEC

Answer: B — UTC (Utilities Technology Council) notification, then 30-day wait



Checkpoint Quiz 4 / 10

If the control link of a remotely controlled station fails, what must happen?

- A. The station must automatically stop transmitting within 3 minutes
- B. Switch to automatic control until the link is restored
- C. The station may continue operating for up to 1 hour
- D. The licensee must notify the FCC immediately

Answer: A — Auto-stop within 3 minutes



Checkpoint Quiz 5 / 10

What is the consequence of fraudulent exam administration by a Volunteer Examiner?

- A. Revocation of VE accreditation only
- B. Suspension of operator license only
- C. Revocation of station license AND suspension of operator license
- D. A written warning from the FCC

Answer: C — BOTH station license revoked AND operator license suspended



Checkpoint Quiz 6 / 10

What is the primary advantage of circular polarization on a ground station antenna when working satellites?

- A. Greater signal bandwidth
- B. Resistance to multipath propagation
- C. Mitigation of spin modulation from tumbling satellites
- D. Reduced path loss

Answer: C — Reduces spin modulation effects from tumbling satellites



Checkpoint Quiz 7 / 10

Which HF digital mode provides the highest data throughput?

- A. PSK31
- B. FT8
- C. WSPR
- D. PACTOR IV

Answer: D — PACTOR IV



Checkpoint Quiz 8 / 10

What does the VIS code at the beginning of an SSTV transmission identify?

- A. The transmitting station's call sign
- B. The SSTV mode being used
- C. The receiving station's ID
- D. The image resolution in pixels

Answer: B — The SSTV mode (color system and resolution)



Checkpoint Quiz 9 / 10

What is the Cabrillo format used for?

- A. Exchanging QSO data between logging programs
- B. Electronic submission of contest logs
- C. Confirming contacts for DXCC awards
- D. Automatic frequency control in digital modes

Answer: B — Electronic contest log submission



Checkpoint Quiz 10 / 10

What does PRB-1 require of state and local governments?

- A. Exempt all amateur antennas from local zoning entirely
- B. Allow unlimited antenna height in residential areas
- C. Reasonably accommodate amateur radio and not preclude all antenna installations
- D. Require permits for all antennas over 10 feet

Answer: C — Reasonably accommodate; cannot preclude all communications



Before Module 2

Your assignment:

- Drill HamStudy sections E1A through E2E
- Target 80%+ confidence on each section before Module 2
- Focus extra time on E2A (satellites) — most students find it unfamiliar
- HamStudy tracks your weak spots and drills those harder — trust the system

Coming in Module 2: Advanced Propagation & Practices

EME, meteor scatter, aurora, TEP, space weather, test equipment, receiver performance